Ross C. Stewart

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Education

Northeastern University — **Sep 2024 – May 2028 (expected)**
Ph.D., Computer Science (Machine Learning & Bioinformatics); GPA: 4.0 — *Boston, MA*
- Khoury College of Computer Sciences; advised in the Radivojac Lab

University of Miami — **Aug 2021 – May 2024**
B.S., Computer Science, Minor in Mathematics; GPA: 3.93, Cum Laude — *Coral Gables, FL*
- Phi Beta Kappa Honor Society
- Top Undergraduate Project Award, Department of Computer Science (2024)

Research Experience

Research Assistant, Khoury College of Computer Sciences — **Sep 2024 – Present**
Radivojac Lab, Northeastern University — *Boston, MA*
- Develop graph attention network models to predict interaction-specific protein–protein interaction perturbations caused by missense variants, outperforming prior state-of-the-art methods (**MutPred-PPI**; published at RECOMB 2026)
- Build skew-normal mixture model methods to calibrate high-throughput functional assay data for clinical variant interpretation (**ExCALIBR**); reclassified 75% of 16,115 variants of uncertain significance as pathogenic or benign with <1% error
- Train models across 30+ residue-level property prediction tasks to uncover mechanisms of pathogenicity
- Awarded NSF ACCESS Explore allocation (MED260059) to support ExCALIBR development

Undergraduate Research Assistant, Dr. Zheng Wang's Bioinformatics Lab — **May 2022 – May 2024**
University of Miami — *Coral Gables, FL*
- Developed deep 3D residual neural network models to refine AlphaFold2 protein tertiary structure predictions (**PICNIC**; published at IEEE BIBM 2024)
- Participated in the CASP15 structure prediction assessment

Other Professional Experience

AI/ML Intern — **Jun 2024 – Aug 2024**
STEMperts — *Houston, TX*
- Engineered LLM pipelines to tailor mathematics education to individual student interests; developed MVTs to demonstrate specific use cases

XR Programmer, University of Miami Innovate Team — **Jan 2022 – May 2022**
University of Miami — *Coral Gables, FL*
- Built XR applications with C#, Unity, and Magic Leap in a scrum-based development team

VR Developer Intern — **May 2022 – Aug 2022**
The Singular Factory Group, LLC — *Miami, FL*
- Built metaverse and VR/AR/MR solutions in C# and Unity, including HoloGIS, a multi-user virtual conferencing tool

Teaching Experience

Teaching Assistant, Machine Learning (graduate) — **Sep 2026 – Dec 2026**
- Northeastern University, Khoury College of Computer Sciences

Teaching Assistant, Computer Science — **Aug 2022 – May 2024**

- University of Miami; instructed 200+ students in data structures, algorithms, machine learning, and bioinformatics; held office hours and graded/designed project assignments

Publications

Abderrazzaq, H., **Stewart, R.**, Tejura, M., McEwen, A.E., Casadei, S., Snyder, M.W., Smith, N., Rubin, A.F., Fowler, D.M., Starita, L.M., Radivojac, P. Classification of In-frame Indels at Scale Using Calibrated Functional and Computational Evidence. *Pac. Symp. Biocomput.* (2027, in press).

Stewart, R., et al. Predicting interaction-specific protein–protein interaction perturbations by missense variants with MutPred-PPI. *The 30th Annual International Conference on Research in Computational Molecular Biology* (RECOMB 2026).

Tejura, M., Chen, Y., McEwen, A.E., **Stewart, R.**, Sverchkov, Y., Laval, F., et al. (6 co-first authors). A scalable approach to resolving variants of uncertain significance. *bioRxiv* 2026.02.14.705848.

Chen, Y., Fayer, S., Jain, S., Benazouz, M., Sverchkov, Y., Stone, J., Sharma, H., Bergquist, T., **Stewart, R.**, Mooney, S.D., Craven, M., Radivojac, P., Starita, L.M., Fowler, D.M., Pejaver, V. Gene- and domain-aware calibration increases the clinical utility of variant effect predictors. *bioRxiv* 2026.02.17.706269.

Biar, C.G., Wang, Z.R., Camp, N.D., Holmes, D.L., Wheelock, M.K., Pendyala, S., McGee, A.V., Gupta, P., McEwen, A.E., Tejura, M., Richardson, M.E., Weyandt, J.D., Coleman, T., **Stewart, R.**, Zeiberg, D., Vandi, A.J., Dawson, S., Radivojac, P., Starita, L.M., Carvill, G.L., James, R.G., Fowler, D.M., Calhoun, J.D. An integrated, scaled approach to resolve TSC2 variants of uncertain significance. *Nat Commun* 17, 8474 (2026).

Casadei, S., Snyder, M., Woo, I., Smith, N., Best, S., Garge, R., Rodriguez-Salas, L., Wenman, C., Seid, O., Hosokai, A., Xu, A., Tejura, M., Gupta, P., Heidl, S., Muffley, L., Ranchalis, J., **Stewart, R.**, Goff, N., Hemker, S., Baierl, J., Dicks, E., Pharoah, P., Bjornkjaer, L., Roenlund, K., Kitzman, J., Bernstein, K., Stergachis, A., Radivojac, P., Fowler, D., Starita, L. Saturation Genome Editing reveals the functional impact of RAD51D and XRCC2 variants. *bioRxiv* 2026.06.12.731983.

Zeiberg, D., **Stewart, R.**, et al. (co-first authors). Gene-based calibration of high-throughput functional assays for clinical variant classification. *bioRxiv* 2025.04.29.651326.

Stewart, R., Zhao, C., et al. (co-first authors). PICNIC: refinement of AlphaFold2 tertiary structure predictions using deep 3D residual neural networks. *IEEE International Conference on Bioinformatics and Biomedicine* (BIBM 2024), Lisbon, Portugal.

Conference Presentations & Invited Talks

- **Talk** – New England Computational Biology (NECB) 2026
- **Talk** – Impact of Genomic Variation on Function (IGVF) Consortium Meeting, 2026
- **Talk** – RECOMB 2026, Thessaloniki, Greece (accepted conference proceedings paper, MutPred-PPI)
- **Talk** – RECOMB-Genetics Satellite Conference, 2026
- **Flash Talk** – Belgrade Bioinformatics Conference (BelBi), 2026 *(Best Flash Talk / Poster Presentation Award)*
- **Poster** – Intelligent Systems for Molecular Biology (ISMB), 2026
- **Poster** – ISCB Congressional Briefing, Washington, D.C., 2026 (presented to congresspeople)
- **Poster** – American Society of Human Genetics (ASHG) Annual Meeting, 2025; **upcoming**, 2026
- **Poster** – Impact of Genomic Variation on Function (IGVF) Consortium Meeting, 2025, 2026
- **Poster** – Critical Assessment of Genome Interpretation (CAGI), 2025

Awards & Honors

- NSF ACCESS Explore Allocation MED260059, 2026: *Gene-based calibration of high-throughput functional assays for clinical variant classification.*
- Best Team Contribution Award, for first-author contributions to scalable VUS resolution, IGVF 2026
- Best Flash Talk / Poster Presentation Award, Belgrade Bioinformatics Conference (BelBi), 2026

- Top Undergraduate Project Award, Department of Computer Science, University of Miami, 2024
- Phi Beta Kappa, University of Miami, 2024
- *Cum Laude*, University of Miami, 2024

Professional Service

- **Reviewer**, Intelligent Systems for Molecular Biology (ISMB), 2025, 2026
- **Reviewer**, Pacific Symposium on Biocomputing (PSB), 2025, 2026
- **Reviewer**, New England Computational Biology (NECB) 2026

Skills

Technical: Python, PyTorch, R, SQL, Git, Docker, Deep Learning, Protein Language Models
Domains: Bioinformatics, Protein Structure, Variant Effect Prediction, Protein–Protein Interactions